

a) Evaluation against the five fundamental properties of an algorithm.

Algorithm A

property

Input

Output

Definiteness

finiteness

Effectiveness

Evaluation

Takes temperature $[1..n]$ and integer n

Produces average.

Each step is clear and unambiguous

loop runs exactly n times and terminates

Uses basic arithmetic operations that are executable.

Algorithm A satisfies all five properties of an algorithm.

Algorithm B

Property

Input

Output

Definiteness

finiteness

Effectiveness

Evaluation

Takes temperature $[1..n]$ and integer n

Returns sum (intended as average)

Steps are clearly defined

Both loops run a finite number of

times

Uses basic arithmetic operations

Algorithm B technically satisfies the five properties, but fails logically (correctness is not a fundamental property, but a practical requirement).

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b) Step-by-step execution with given Input

Input

temperatures = [20, 25, 22, 24, 21]
 $n = 5$

Algo A execution

Step	total
Initialize	0
Add 20	20
Add 25	45
Add 22	67
Add 24	91
Add 21	112

$$\text{average} = \text{total} / n = 112 / 5 = 22.4$$

Output: 22.4

Algo B Execution

first loop (summation)

step	Sum
Initialize	20
Add 25	45
Add 22	67
Add 24	91
Add 21	112

Second loop (division)
loop runs 5 times:

Iteration 1: $\text{sum} = 112 / 5 = 22.4$

Iteration 2: $\text{sum} = 22.4 / 5 = 4.48$

Iteration 3: $\text{sum} = 4.48 / 5 = 0.896$

Iteration 4: $\text{sum} = 0.896 / 5 = 0.1792$

Iteration 5: $\text{sum} = 0.1792 / 5 = 0.03584$

Output: 0.03584

c) Logical error in Algo B

Error:

The algorithm divides the accumulated sum by n repeatedly inside a loop.

The average should be calculated as:

$$\text{Average} = \frac{\sum \text{temperatures}}{n}$$

Algo B incorrectly computes:

$$\frac{\sum}{n^2}$$

violated principle:

Division is not associative or idempotent; repeated division alters the result.

d) Comparison of Algo

Criterion	Algo A	Algo B
Simplicity	Simple and clear	Unnecessarily Complex
Correctness	Correct	Incorrect
Efficiency	$O(n)$	$O(n)$ with redundant work
Reliability	High	low

final choice:-

Algo A is suitable for production because it is correct, efficient, and easy to understand.

e) Test Strategy

Test Case	Input	Expected output
Normal Case	$[20, 25, 22, 24, 21], n=5$	22.4
Single value	$[30], n=1$	30
Zero elements	$[], n=0$	Error/Exception
Negative Values	$[-5, -10, -3], n=3$	-6
Mixed Values	$[10, -10, 20, -20], n=4$	0

Q.2

Worst case for finding a username in a list of 10 million users.

i) for linear search

Worst case = $n = 10$ million Comparisons

ii) Binary Search

$$\log_2(10^7) \approx 24$$

Hash Table

Average case ≈ 1 Comparison

b) Algorithm

Time Complexity

i) Linear Search

$O(n)$

ii) Binary Search

$O(\log n)$

iii) Hash Table

$O(1)$ Average

c)	n	Linear	Binary	Hash
	1000	100	7	1
	10000	1000	10	1
	10000	10000	14	1
	100000	100000	17	1
	1000000	1000000	20	1
	10000000	10000000	24	1

hash remain constant & linear search. Time complexity grow as the value n grow

d) 50,000 searches/sec (1 ns per comparison)

- Linear

$$50,000 \times 10^7 = 5 \times 10^{11} \text{ ns} = 500 \text{ sec}$$

- Binary

$$50,000 \times 24 = 1.2 \text{ ms}$$

- Hash

$$50,000 \times 1 = 0.05 \text{ ms}$$

Only Binary Search & Hash tables can handle the load.

e) Linear Searches $= K \cdot n$

Sort + binarysearch $n \log n + K \log n$

Sorting is worth when $n \log n < K \log n$

for large K (many searches),
Binary Search wins.

3.

a Time Complexity

Implementation A

i) Nested loops $\rightarrow O(n \cdot m)$

• Implementation B

ii) Hash creation $O(n)$ + lookup $O(m)$

$\rightarrow O(n+m)$

b) Implementation	space
A	$O(1)$
B	$O(n)$

c) Given $n=40$, $m=5$, 100,000 students

Implementation A:

$40 \times 5 \times 100,000 = 20,000,000$ comparisons/day.

Implementation B:

$40 \times 8 = 320$ bytes per student

d) Each comparison = $0.1 \mu s$

Implementation A time

$$200 \times 0.1 = 20 \mu s$$

memory for B

$$10,000 \times 320 = 3.2 \text{ MB}$$

Implementation B is better (faster + memory safe)

2) Hybrid Approach:

if $m < 2 \rightarrow$ use nested loop

if $m \geq 10 \rightarrow$ use hash set

Switch threshold around $m = 5$

4) ⑨ Big-O pr.

b) Data type	Best Algo	Reason
Nearly sorted	Insertion sort	$O(n)$ best case
Random	merge / Quick	$O(n \log n)$
Reverse Sorted	merge sort	$O(n \log n)$

c) Θ vs O Explanation

$\Theta(n \log n)$: Tight bound (best = worst),

$O(n \log n)$: Upper bound only

merge sort is predictable, Quick sort is not.

d) Decision Guide

- i) Small ($n < 50$): Insertion sort
- ii) medium Random: Quick Sort
- iii) Large or critical: merge sort
- iv) Reverse Sorted: merge sort

e) $T_n = 5n^2 + 100n + 1000 < 10n \log_2 n$

for small $n (< 50)$: Hy story sort maybe faster

for large n : n^2 dominates $\rightarrow$ merge sort faster.